

3.6 Problem Set: Mechanical Advantage — Solutions

1. Pulleys.

- (a) $MA = 1$. A fixed pulley does not multiply force; it only changes the *direction* of the effort (pull down to lift up).
- (b) Two supporting strands $\Rightarrow MA = 2$, so $F_{\text{in}} = \frac{240}{2} = \mathbf{120}$ N.
- (c) $MA = 4$: $F_{\text{in}} = \frac{600}{4} = \mathbf{150}$ N. Rope pulled $= 4 \times 0.50 = \mathbf{2.0}$ m (force divided by 4, distance multiplied by 4).

2. Lever.

- (a) $MA = \frac{\text{effort arm}}{\text{load arm}} = \frac{1.20}{0.20} = \mathbf{6}$.
- (b) $F_{\text{in}} = \frac{F_{\text{out}}}{MA} = \frac{850}{6} \approx \mathbf{142}$ N.
- (c) Rock rises $\frac{0.30}{6} = 0.05$ m = **5** cm. Trade-off: you gain force ($6\times$) but lose distance ($6\times$) — work in = work out.

3. Chain and stuck car.

- (a) $T = \frac{F_{\text{pull}}}{2 \sin \theta} = \frac{300}{2 \sin 10^\circ} = \frac{300}{2(0.1736)} \approx \mathbf{864}$ N.
- (b) $MA = \frac{1}{2 \sin 10^\circ} = \frac{1}{0.347} \approx \mathbf{2.88}$ (matches $864/300$).
- (c) As $\theta \rightarrow 0$, $\sin \theta \rightarrow 0$ and the force on the car $\rightarrow \infty$ — a small sideways pull on a nearly straight chain produces an enormous tension. Practical danger: the chain (or attachment) can snap and whip back; the force can far exceed the chain's rating.

4. Hydraulic lift.

- (a) Small: $r = 1.0$ cm, $A = \pi(1.0)^2 \approx \mathbf{3.14}$ cm². Large: $r = 15$ cm, $A = \pi(15)^2 \approx \mathbf{707}$ cm².
- (b) $MA = \frac{A_{\text{large}}}{A_{\text{small}}} = \frac{707}{3.14} = \mathbf{225}$ (equivalently $(15/1)^2 = 225$).
- (c) $F_{\text{out}} = 225 \times 50 = \mathbf{1.13 \times 10^4}$ N (11,250 N, $\approx$ the weight of an 1150 kg car).

$$(d) \ 1500 \text{ PSI} \times \frac{6.895 \text{ kPa}}{1 \text{ PSI}} = \mathbf{1.03 \times 10^4 \text{ kPa}} \ (10,340 \text{ kPa} = 10.3 \text{ MPa}).$$